biowatt

BioWatt MFC energy-harvester board: manufacturing manual

> **Status: design only; nothing built or tested with a real fuel cell.** PCB: **order-ready: every automatic check passes**. Setting resistors derived from the TI BQ25570 datasheet equations (SLUSBH2G).

1. What is made

A 44 x 40 mm board: BQ25570 harvester (boost charger with MPPT at 50 % of Voc, nano-power buck) from a microbial fuel cell input (J1) into a protected single Li-ion/LiPo cell (J2), with a regulated 3.35 V output and a VBAT_OK flag on J3. Thirteen 1 % megohm resistors set VBAT_OV 3.95 V, VBAT_OK 3.35 V falling / 3.63 V rising and VOUT 3.35 V; the region around them has no ground pour.

2. Deliverables

Item	Where	State
PCB	`utility10/pcb/release/biowatt/` and `biowatt/hardware/biowatt/`	**order-ready: every automatic check passes**
Energy budget model	`biowatt/sim/budget.py`, `biowatt/docs/budget.md`	assumption-based

3. Bill of materials

Ref	Value	Package	Part (manufacturer)	LCSC	Assembly
U1	BQ25570RGRR	QFN-20-1EP_3.5x3.5mm_P0.5mm_EP2x2mm	BQ25570RGRR	C506250	SMT (JLC)
J1	MFC in (+ / -)	JST_PH_B2B-PH-K_1x02_P2.00mm_Vertical	JST B2B-PH-K-S	C131337	hand/THT
J2	Storage cell (+ / -), protected	JST_PH_B2B-PH-K_1x02_P2.00mm_Vertical	JST B2B-PH-K-S	C131337	hand/THT
J3	Tag out (VOUT, GND, VBAT_OK)	JST_PH_B3B-PH-K_1x03_P2.00mm_Vertical	JST B3B-PH-K-S	C131339	hand/THT
L1	22u	L_Coilcraft_LPS4018	Coilcraft LPS4018-223MRC	C5278935	SMT (JLC)
L2	10u	L_Changjiang_FNR3015S	ZE ZENR3015T100M	C42427989	SMT (JLC)
C1	4.7u	C_0603_1608Metric	Samsung CL10A475KP8NNNC X5R 10V	C1705	SMT (JLC)
C2	100n	C_0603_1608Metric	Samsung CL10B104KB8NNNC 0603 100nF X7R 50V	C1591	SMT (JLC)

Ref	Value	Package	Part (manufacturer)	LCSC	Assembly
C3	4.7u	C_0603_1608Metric	Samsung CL10A475KP8NNNC X5R 10V	C1705	SMT (JLC)
C4	22u/10V	C_0805_2012Metric	Samsung CL21A226MPQNNNE	C29277	SMT (JLC)
C5	22u/10V	C_0805_2012Metric	Samsung CL21A226MPQNNNE	C29277	SMT (JLC)
C6	10n	C_0603_1608Metric	Samsung CL10B103KB8NNNC 0603 10nF X7R 50V	C1589	SMT (JLC)
R3	5.6M	R_0603_1608Metric	FOJAN FRC0603F5604TS	C2933237	SMT (JLC)
R2	1M	R_0603_1608Metric	UNI-ROYAL 0603WAF1004T5E	C22935	SMT (JLC)
R1	5.6M	R_0603_1608Metric	FOJAN FRC0603F5604TS	C2933237	SMT (JLC)
R4	4.3M	R_0603_1608Metric	FH RS-03L4304FT	C136595	SMT (JLC)
R5	5.6M	R_0603_1608Metric	FOJAN FRC0603F5604TS	C2933237	SMT (JLC)
R6	1M	R_0603_1608Metric	UNI-ROYAL 0603WAF1004T5E	C22935	SMT (JLC)
R7	1M	R_0603_1608Metric	UNI-ROYAL 0603WAF1004T5E	C22935	SMT (JLC)
R8	1M	R_0603_1608Metric	UNI-ROYAL 0603WAF1004T5E	C22935	SMT (JLC)
R9	1M	R_0603_1608Metric	UNI-ROYAL 0603WAF1004T5E	C22935	SMT (JLC)
R10	4.3M	R_0603_1608Metric	FH RS-03L4304FT	C136595	SMT (JLC)
R11	5.6M	R_0603_1608Metric	FOJAN FRC0603F5604TS	C2933237	SMT (JLC)
R12	1M	R_0603_1608Metric	UNI-ROYAL 0603WAF1004T5E	C22935	SMT (JLC)
R13	1M	R_0603_1608Metric	UNI-ROYAL 0603WAF1004T5E	C22935	SMT (JLC)

Bought-in

Protected single Li-ion/LiPo cell (3.7 V) with JST-PH 2 lead; MFC with JST-PH 2 lead; load tag on J3 (BLE/LoRa class: average current well under 1 mA).

4. Make the board

1. Upload `utility10/pcb/release/biowatt/1_fabrication/biowatt_gerbers.zip` to the PCB fab (2 layers, 1.6 mm, ENIG or lead-free HASL, green mask, white silk; 1 oz is fine). 2. For assembly upload `2_assembly/biowatt_JLC_BOM.csv` and `biowatt_JLC_CPL.csv`; check every polarity in the preview (diodes, electrolytic capacitor, USB-C, ESP32 module pin 1). 3. Through-hole connectors and pin sockets are listed with LCSC numbers but are normally hand-soldered. 4. General rules, tolerances and inspection:

`manuals/STANDARD_PROCEDURES.md` sections A and B. The megohm resistors are 0603 1 % parts (5.6 M, 4.3 M, 1 M): check the CPL preview and clean flux residue thoroughly afterwards, then keep the board dry.

5. Commissioning

1. Bench supply 0.8 V, current-limited 5 mA, on J1: with no cell fitted VSTOR (TP2) rises. 2. Fit a protected cell: VBAT_OK goes high above about 3.63 V and drops below 3.35 V. 3. VOUT (TP3) reads 3.35 V. 4. Then connect the MFC.

6. Limits

Cold start needs VIN of at least 600 mV: a single MFC (0.5-0.7 V) is marginal, use two in series or a start-up source. C6 is a stocked X7R 10 nF where the datasheet asks for a low-leakage part. MFC power (about 20 uW) supports only a BLE/LoRa tag. Not tested on hardware.